An Agricultural Prediction and Decision-Support System

A Project Report Submitted in the partial fulfillment of the requirements for the award of the degree of

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In

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Declaration

The Project Report entitled **"AgriIntelligence: Agricultural Analytics and Decision-Support System"** is a record of bonafide work carried out by **Kasi Sai Akash (2420030770) and Sai Shreeja Tedla (2420030631)**, submitted in partial fulfilment of the requirements for the award of the degree of **Bachelor of Technology (B.Tech) in Computer Science and Engineering** at **K L Deemed to be University**.

The project focuses on integrating agricultural crop-yield, weather, and market-price data to develop a machine-learning-based agricultural analytics and decision-support platform. The system incorporates data preprocessing, exploratory data analysis, Random Forest-based yield and price prediction, weather-risk assessment, and crop recommendation through an interactive Streamlit dashboard.

We hereby declare that the work presented in this report is our own work carried out under the guidance of **Ms. B. Prathusha, Assistant Professor, Department of Computer Science and Engineering**, and that the results embodied in this report are original.

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CERTIFICATE

This is to certify that the project report entitled **"AgriIntelligence: An Integrated Agricultural Analytics and Decision-Support System"** is a bonafide work done and submitted by **Kasi Sai Akash (2420030770) and Sai Shreeja Tedla (2420030631)** in partial fulfilment of the requirements for the award of the degree of **BACHELOR OF TECHNOLOGY in Computer Science and Engineering**, Department of Computer Science and Engineering, K L (Deemed to be University), during the academic year **2026–2027**.

Signature of the Guide

Signature of the Course Coordinator **Signature of the HOD**

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Abstract

Agriculture plays a significant role in the economy and food security of many countries. However, farmers face several challenges while selecting suitable crops, estimating crop production, understanding market-price fluctuations, and managing weather-related risks. Traditional agricultural decisions often depend on previous experience and limited information, which may lead to uncertainty and financial losses.

AgriIntelligence is an integrated agricultural prediction and decision-support system developed to assist users in making informed agricultural decisions through data analysis and machine learning techniques. The system combines agricultural, weather, and market-related information to provide multiple analytical features through a single interactive dashboard.

The project includes data loading, preprocessing, data integration, exploratory data analysis, and feature engineering to prepare the datasets for machine learning. Random Forest algorithms are used as the primary modelling approach for crop-yield and market-price prediction. The system also includes a crop recommendation module to suggest suitable crops based on available agricultural parameters and a weather-risk analysis module that evaluates conditions such as rainfall, temperature, and humidity.

The application is developed using Python, Pandas, NumPy, Scikit-learn, and Streamlit. The dashboard provides interactive visualizations, prediction results, historical trends, and agricultural insights in an understandable format.

The main objective of AgriIntelligence is to bring multiple agricultural analytical functions together in one platform, helping users understand agricultural conditions, explore predictions, and make more informed crop-planning and market-related decisions.

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CHAPTER 1

Introduction

1.1 Background of the Project

Agriculture is one of the most important sectors, providing food, employment, and raw materials for various industries. Agricultural productivity is influenced by several factors, including soil conditions, rainfall, temperature, humidity, crop selection, and market demand.

Farmers often face difficulties in identifying suitable crops, estimating expected yield, predicting market prices, and understanding weather conditions. These challenges can affect agricultural productivity and profitability.

With the development of data science and machine learning, agricultural datasets can be analysed to identify patterns and generate predictions. Machine learning techniques help process historical information and provide insights that can support agricultural planning.

AgriIntelligence is developed to bring these capabilities together in a single platform. It combines crop recommendation, yield prediction, price prediction, and weather-risk analysis through an interactive dashboard.

1.2 Motivation

The motivation behind developing AgriIntelligence is to address the difficulties faced in agricultural planning and decision-making.

Some of the major challenges include:

- Difficulty in selecting suitable crops based on agricultural conditions.
- Uncertainty in estimating crop yield before harvesting.
- Fluctuations in agricultural market prices.
- Weather conditions that can negatively affect crop production.
- Lack of a unified platform to analyse multiple agricultural factors.

These challenges motivated us to develop an integrated system that uses historical agricultural data and machine learning techniques to generate useful predictions and insights.

1.3 Problem Statement

Agricultural decisions depend on several interconnected factors, including environmental conditions, crop characteristics, historical production, and market prices.

However, analysing these factors separately makes it difficult to obtain a complete understanding of agricultural conditions. Farmers and other users may not have access to an integrated analytical platform that provides crop recommendations, yield estimates, price predictions, and weather-risk insights.

Therefore, AgriIntelligence is developed to analyse agricultural datasets using data-processing and machine-learning techniques and present the results through an interactive dashboard.

1.4 Objectives

The main objective of AgriIntelligence is to develop an integrated agricultural decision-support platform using data analytics and machine learning.

The specific objectives are:

1. To collect, load, and understand agricultural datasets.
2. To clean and preprocess the datasets by handling missing values, duplicates, and inconsistent data.
3. To perform exploratory data analysis to identify agricultural trends and patterns.
4. To apply feature engineering to prepare relevant variables for machine learning.
5. To develop a crop recommendation module based on agricultural input parameters.
6. To predict crop yield using a Random Forest machine-learning model.
7. To analyse historical agricultural prices and predict future or target market prices.
8. To evaluate weather-related risks using rainfall, temperature, and humidity information.
9. To evaluate prediction performance using appropriate model evaluation metrics.
10. To develop an interactive Streamlit dashboard that presents predictions and insights in an understandable manner.

1.5 Scope of the Project

The scope of AgriIntelligence includes:

- Agricultural dataset processing and analysis.
- Crop recommendation based on available input parameters.
- Crop-yield prediction using machine learning.
- Agricultural market-price prediction.
- Weather-risk analysis.
- Interactive charts and visualizations.
- Presentation of agricultural insights through a centralized dashboard.

The current implementation focuses on historical datasets and the machine-learning models developed using those datasets. Real-time data integration and large-scale deployment can be considered in future versions.

1.6 Importance of the Application

AgriIntelligence provides several benefits:

- Helps users explore suitable crop choices.
- Provides estimated crop-yield information.
- Helps understand agricultural price patterns.
- Highlights potential weather-related risks.
- Reduces the need to analyse multiple datasets manually.
- Combines different agricultural analytical modules in one application.
- Presents machine-learning results through an easy-to-use interface.

1.7 Target Users / Audience

The intended users of AgriIntelligence include:

1. **Farmers:** To explore crop recommendations and agricultural predictions.
2. **Agricultural Students:** To understand the application of machine learning in agriculture.
3. **Agricultural Researchers:** To analyse agricultural trends and environmental relationships.
4. **Agricultural Analysts:** To study yield, price, and weather-related patterns.
5. **Educational Institutions:** To demonstrate data science and machine learning applications in agriculture.

CHAPTER 2

Software Requirement

SYSTEM REQUIREMENTS

2.1 Hardware Requirements

The following hardware configuration is suitable for developing and running the application.

Component	*Requirement*
Processor	*Intel Core i3 or above*
RAM	*Minimum 4 GB; 8 GB recommended*
Storage	*Minimum 2 GB available space*
Operating System	*Windows 10/11 or compatible OS*
Internet	*Required for downloading libraries and accessing online resources*

2.2 Software Requirements

Software	*Purpose*
Python	*Main programming language*
Visual Studio Code	*Development environment*
Streamlit	*Interactive dashboard development*
Pandas	*Dataset loading and processing*
NumPy	*Numerical calculations*
Scikit-learn	*Machine-learning model development*
Matplotlib	*Data visualization*
Seaborn	*Exploratory data visualization*
Git	*Version control*
GitHub	*Source-code hosting*

Web Browser ***Accessing the dashboard***

2.3 Development Tools and Frameworks

2.3.1 Python

Python is the primary programming language used in AgriIntelligence. It provides a wide range of libraries for data processing, statistical analysis, visualization, and machine learning.

2.3.2 Pandas

Pandas is used to load, manipulate, clean, and analyse agricultural datasets. It helps handle missing values, duplicate records, data types, and tabular data operations.

2.3.3 NumPy

NumPy is used for numerical calculations and mathematical operations. It supports efficient processing of numerical agricultural variables.

2.3.4 Scikit-learn

Scikit-learn is used to develop and evaluate machine-learning models. It provides tools for data preprocessing, model training, prediction, and evaluation.

The Random Forest algorithm is used as the primary model for the project's prediction tasks where applicable.

2.3.5 Streamlit

Streamlit is used to develop the interactive web dashboard. It allows Python-based analytical results and machine-learning predictions to be presented through an easy-to-use interface.

2.3.6 Matplotlib and Seaborn

These visualization libraries are used to create graphs and charts that help identify relationships, trends, distributions, and patterns in agricultural datasets.

2.3.7 Visual Studio Code

Visual Studio Code is used to write, edit, debug, and execute the Python scripts and Streamlit application.

2.3.8 Git and GitHub

Git is used for version control, while GitHub is used to host and manage the project's source code.

CHAPTER 3

SYSTEM DESIGN AND ARCHITECTURE

3.1 System Architecture

AgriIntelligence follows a data-processing and machine-learning architecture

consisting of the following major layers:

1. Data Input Layer

Agricultural datasets containing crop, yield, weather, location, and market-price information are loaded into the system.

2. Data Preprocessing Layer

The input datasets are cleaned and prepared by handling missing values, duplicates, inconsistent formats, and other data-quality issues.

3. Exploratory Data Analysis Layer

Statistical summaries, graphs, and visualizations are generated to understand the datasets and identify important patterns.

4. Feature Engineering Layer

Relevant features are selected, transformed, and prepared for machine-learning models.

5. Machine Learning Layer

The system uses machine-learning techniques for crop recommendation, yield prediction, and market-price prediction. Weather-related information is analysed separately to identify potential risks.

6. Dashboard and Output Layer

The results are displayed through an interactive Streamlit dashboard containing prediction outputs, charts, tables, and agricultural insights.

3.2 System Workflow

The overall workflow of AgriIntelligence is as follows:

1. ***Load agricultural datasets.***
2. ***Understand dataset structure and variables.***
3. ***Perform data cleaning and preprocessing.***
4. ***Integrate relevant datasets.***
5. ***Perform exploratory data analysis.***
6. ***Apply feature engineering.***
7. ***Train and evaluate machine-learning models.***
8. ***Generate crop, yield, and price predictions.***
9. ***Perform weather-risk analysis.***
10. ***Display results through the Streamlit dashboard.***

3.3 Data Flow Diagram (DFD)

A Data Flow Diagram represents how information moves through the application.

In AgriIntelligence, the user provides or selects agricultural information. The system processes the datasets, performs machine-learning analysis, and displays the results.

DFD Components:

- ***External Entity: User***
- ***Input: Agricultural datasets and prediction parameters***
- ***Processes: Data preprocessing, EDA, feature engineering, model prediction and weather-risk analysis***
- ***Output: Crop recommendations, yield estimates, price predictions and weather-risk insights***

3.4 Module Description

The application consists of the following major modules:

1. ***Data Understanding and Loading***
2. ***Data Cleaning and Preprocessing***
3. ***Data Integration***
4. ***Exploratory Data Analysis***
5. ***Feature Engineering***
6. ***Crop Recommendation***
7. ***Crop Yield Prediction***
8. ***Market Price Prediction***
9. ***Weather Risk Analysis***
10. ***Streamlit Dashboard***

CHAPTER 4

IMPLEMENTATION AND USER INTERFACE

4.1 Data Loading and Understanding

The first stage of the project involves loading the agricultural datasets and understanding their structure.

The datasets are examined to identify:

- ***Number of rows and columns.***
- ***Column names and data types.***
- ***Missing values.***
- ***Duplicate records.***
- ***Statistical summaries.***

- ***Important agricultural variables.***

This stage helps understand the available information before performing further processing.

4.2 Data Cleaning and Preprocessing

Data preprocessing is performed to improve dataset quality and prepare the information for analysis.

The operations include:

- ***Handling missing values.***
- ***Removing duplicate records where appropriate.***
- ***Correcting data types.***
- ***Handling inconsistent values.***
- ***Preparing numerical and categorical variables.***

The cleaned datasets are used in subsequent analysis and machine-learning stages.

4.3 Data Integration

Data integration combines relevant agricultural information from different datasets.

The project uses relevant crop, weather, location, yield, and market-price information to support the different analytical modules.

The purpose of this stage is to create suitable datasets for prediction and analysis.

4.4 Exploratory Data Analysis (EDA)

Exploratory Data Analysis is performed to understand the distribution and relationships of agricultural variables.

The analysis includes:

- ***Statistical summaries.***
- ***Crop-wise comparisons.***
- ***Weather-condition analysis.***
- ***Yield distribution.***
- ***Market-price trends.***
- ***Correlation analysis.***
- ***Graphical visualization.***

EDA helps identify patterns and relationships that may influence agricultural outcomes.

4.5 Feature Engineering

Feature engineering involves selecting and preparing relevant input variables for

machine-learning models.

It helps improve the usefulness of the input data by:

- ***Selecting relevant features.***
- ***Transforming numerical and categorical variables.***
- ***Preparing model input data.***
- ***Applying appropriate preprocessing techniques.***

The selected features depend on the requirements of each prediction module.

4.6 Crop Recommendation Module

The Crop Recommendation module provides crop suggestions based on the agricultural parameters supported by the dataset and the implemented recommendation logic.

The module analyses available input information and generates suitable crop suggestions.

The output is presented through the dashboard to help users explore potential crop choices.

4.7 Crop Yield Prediction Module

Crop Yield Prediction estimates agricultural production using historical agricultural information.

The project uses the Random Forest algorithm for yield prediction.

Random Forest combines predictions from multiple decision trees to produce a final result. It can model nonlinear relationships between agricultural input variables and yield.

The general implementation process includes:

1. ***Preparing the yield dataset.***
2. ***Selecting relevant input features.***
3. ***Splitting the data into training and testing sets.***
4. ***Training the Random Forest model.***
5. ***Generating predictions.***
6. ***Evaluating the model performance.***

The model performance can be assessed using:

- ***Mean Absolute Error (MAE).***
- ***Mean Squared Error (MSE).***
- ***Root Mean Squared Error (RMSE).***
- ***R-squared (R^2).***

4.8 Market Price Prediction Module

The Market Price Prediction module analyses historical agricultural market-price information to estimate crop prices.

The system uses the available historical price data and relevant features to train a prediction model.

The implementation includes:

1. ***Loading historical market-price data.***
2. ***Cleaning and preparing the dataset.***
3. ***Selecting relevant prediction features.***
4. ***Training the price-prediction model.***
5. ***Generating estimated prices.***
6. ***Comparing actual and predicted values.***

The dashboard presents the prediction results and historical price information to help users understand market trends.

4.9 Weather Risk Analysis Module

Weather conditions play an important role in agricultural production.

The Weather Risk Analysis module evaluates available weather information such as:

- ***Rainfall.***
- ***Temperature.***
- ***Humidity.***

The system analyses these parameters to identify potentially unfavourable weather conditions and display weather-related risk insights.

The module helps users understand environmental conditions that may influence agricultural activities.

4.10 Streamlit Dashboard

The Streamlit dashboard acts as the main interface of AgriIntelligence.

It brings the different analytical modules together in a single application.

The dashboard provides:

- ***Agricultural data insights.***
- ***Crop recommendations.***
- ***Crop-yield predictions.***
- ***Market-price predictions.***
- ***Weather-risk analysis.***

- ***Interactive charts and visualizations.***
- ***Prediction summaries.***

4.10.1 Dashboard Overview

The overview section provides a general introduction to the platform and gives users access to the different analytical modules.

Fig. 4.1: AgriIntelligence Dashboard Overview

4.10.2 Crop Recommendation Page

This page presents crop suggestions based on the available agricultural input parameters.

Fig. 4.2: Crop Recommendation Module

4.10.3 Crop Yield Prediction Page

This page allows users to view the estimated crop yield generated by the trained machine-learning model.

Fig. 4.3: Crop Yield Prediction Module

4.10.4 Market Price Prediction Page

This page displays estimated market prices and relevant historical price information.

Fig. 4.4: Market Price Prediction Module

4.10.5 Weather Risk Analysis Page

This page presents weather-related information and identifies potential risks based on the available weather variables.

Fig. 4.5: Weather Risk Analysis Module

4.10.6 Data Visualization Page

This section displays graphical representations of agricultural datasets and prediction results to make the information easier to understand.

Fig. 4.6: Agricultural Data Visualization

CHAPTER 5

Conclusion and Future Aspects

CHAPTER 5
TESTING, RESULTS AND FUTURE ENHANCEMENTS

5.1 Testing

Testing is performed to verify that the different modules of AgriIntelligence work as expected. The following testing activities are considered:

1. Data Loading Testing

Verify that agricultural datasets are loaded correctly.

2. Data Preprocessing Testing

Check whether missing values, duplicate records, and inconsistent data are handled appropriately.

3. Model Testing

Verify that trained machine-learning models generate predictions using valid input data.

4. Dashboard Testing

Check whether dashboard components, input controls, charts, and prediction outputs work correctly.

5. Integration Testing

Verify that data-processing modules, prediction models, and the Streamlit dashboard work together.

5.2 Model Evaluation

The performance of the prediction models is evaluated using suitable evaluation metrics.

Mean Absolute Error (MAE)

Measures the average absolute difference between actual and predicted values.

Mean Squared Error (MSE)

Measures the average squared difference between actual and predicted values.

Root Mean Squared Error (RMSE)

Represents the square root of the mean squared prediction error.

R-squared (R^2)

Indicates how much variation in the target variable is explained by the model.

The actual evaluation results obtained from the implemented models should be included in this section.

5.3 Results

The developed AgriIntelligence platform provides an integrated interface for agricultural analysis and prediction.

The system successfully brings together the implemented modules for crop recommendation, crop-yield prediction, market-price prediction, and weather-risk analysis.

The dashboard presents the outputs through visualizations, summaries, and prediction results.

The results demonstrate how data analytics and machine-learning techniques can be applied to agricultural decision-support applications.

5.4 Challenges Faced During Development

The following challenges were encountered or considered during development:

1. **Dataset Availability:** Finding suitable datasets containing relevant agricultural

information.

2. **Data Preprocessing:** Handling missing values, duplicates, and inconsistent formats.
3. **Data Integration:** Combining different agricultural datasets with appropriate features.
4. **Feature Selection:** Identifying useful input variables for each prediction module.
5. **Model Selection:** Choosing an appropriate machine-learning algorithm for structured agricultural data.
6. **Prediction Accuracy:** Evaluating prediction errors and understanding the differences between actual and predicted values.
7. **Dashboard Integration:** Connecting the different analytical modules within one Streamlit application.
8. **Deployment:** Managing Python dependencies and ensuring compatibility between the local development environment and the hosting environment.

5.5 Limitations

Although AgriIntelligence provides multiple agricultural analytical features, the current implementation has certain limitations:

1. **Dataset Dependency:** Prediction quality depends on the availability, accuracy, and completeness of the historical datasets.
2. **Limited Real-Time Data:** The current system primarily relies on available datasets rather than continuous real-time agricultural updates.
3. **Geographical Coverage:** Predictions may not generalize equally to every geographical region because agricultural conditions vary by location.
4. **Model Limitations:** Random Forest predictions depend on the training data and may not capture every unexpected agricultural or market change.
5. **Weather Uncertainty:** Unexpected weather events may not be fully represented by historical weather information.
6. **Market Fluctuations:** Agricultural prices are influenced by demand, supply, transportation, government policies, and other external factors that may not all be included in the dataset.
7. **Recommendation Scope:** Crop recommendations are limited to the crops and input parameters supported by the available data and implemented logic.

5.6 Solutions and Improvements Implemented

To address development challenges, the following measures were incorporated:

- Data cleaning and preprocessing to improve input quality.
- Data integration to combine relevant agricultural information.
- Feature engineering to prepare suitable model inputs.
- Random Forest modelling for structured prediction tasks.
- Evaluation using appropriate regression metrics.
- Weather-risk analysis to consider environmental conditions.
- A centralized Streamlit dashboard to present multiple outputs together.

5.7 Future Enhancements

The following improvements can extend the capabilities of AgriIntelligence:

1. **Real-Time Weather Integration:** Connect weather APIs to provide current conditions and forecasts.
2. **Soil Health Analysis:** Include soil pH, nitrogen, phosphorus, potassium, and soil-moisture information.
3. **Satellite Data Integration:** Use satellite imagery and remote sensing for crop monitoring.
4. **Advanced Machine Learning:** Compare Random Forest with XGBoost, Gradient

Boosting, and deep-learning models.

5. **Profit Prediction:** Estimate expected revenue and profit by combining yield estimates with market prices and production costs.
6. **Geographical Mapping:** Add interactive maps showing regional crop suitability, yield, prices, and weather risks.
7. **Mobile Application:** Develop a mobile-friendly version to improve accessibility.
8. **Explainable AI:** Provide explanations for crop recommendations and predictions by identifying influential input features.

5.8 Conclusion

AgriIntelligence was developed as an integrated agricultural prediction and decision-support platform using data analytics and machine-learning techniques.

The project includes the major stages of an agricultural data-science workflow, beginning with dataset loading and preprocessing, followed by data integration, exploratory data analysis, and feature engineering.

The system incorporates crop recommendation, crop-yield prediction, market-price prediction, and weather-risk analysis. Random Forest is used as the primary machine-learning approach for the relevant prediction tasks because it can capture nonlinear relationships in structured agricultural data and provides a practical starting point for model development.

The Streamlit dashboard brings the implemented modules together and presents their results through interactive visualizations and prediction outputs.

Overall, the project demonstrates the application of data science and machine learning in agriculture. It provides a foundation for developing more advanced agricultural decision-support systems using real-time information, additional environmental variables, and improved prediction techniques.

CHAPTER 6
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6.1 Research Papers

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[4] "Predicting Prices of Staple Crops Using Machine Learning: A Systematic Review of Studies on Wheat, Corn, and Rice," *Sustainability*, 2025.

6.2 Technical Documentation

1. Python Official Documentation – https://docs.python.org/3/
2. Pandas Documentation – https://pandas.pydata.org/docs/
3. NumPy Documentation – https://numpy.org/doc/
4. Scikit-learn Documentation – https://scikit-learn.org/stable/
5. Streamlit Documentation – https://docs.streamlit.io/
6. Matplotlib Documentation – https://matplotlib.org/stable/
7. Seaborn Documentation – https://seaborn.pydata.org/
8. Git Documentation – https://git-scm.com/doc
9. GitHub Documentation – https://docs.github.com/

CHAPTER 7

APPENDICES

7.1 Project Directory Structure

The project is organized into different directories to separate datasets, processing scripts, machine-learning modules, and the dashboard.

Example structure:

AgriIntelligence/

```
├── data/
│   ├── raw/
│   └── processed/
│
├── src/
│   ├── 01_data_understanding.py
│   ├── 02_data_cleaning.py
│   ├── 03_data_integration.py
│   ├── 04_eda.py
│   ├── 05_feature_engineering.py
│   ├── 06_yield_prediction.py
│   ├── 07_price_prediction.py
│   ├── 08_weather_risk.py
│   └── 09_crop_recommendation.py
│
├── dashboard/
│   └── app.py
│
├── requirements.txt
└── README.md
```

Note: Adjust this directory structure to match the final version of your actual repository.

7.2 Sample Random Forest Implementation

The following illustrates the general Random Forest regression approach used for prediction.

```
from sklearn.ensemble import RandomForestRegressor
from sklearn.metrics import (
    mean_absolute_error,
    mean_squared_error,
    r2_score
)

model = RandomForestRegressor(
    n_estimators=100,
    random_state=42
)

model.fit(X_train, y_train)

predictions = model.predict(X_test)

mae = mean_absolute_error(y_test, predictions)
```

```
mse = mean_squared_error(y_test, predictions)
rmse = mse ** 0.5
r2 = r2_score(y_test, predictions)

print("MAE:", mae)
print("RMSE:", rmse)
print("R2 Score:", r2)
```

The code is an illustrative implementation. Use the exact hyperparameters and preprocessing steps from the final project code when submitting.

7.3 Running the Application

The application can be executed using the following commands from the project root directory.

```
# Activate virtual environment
.venv\Scripts\activate

# Install dependencies
pip install -r requirements.txt

# Run the Streamlit application
streamlit run dashboard/app.py
```

7.4 Screenshots

Include the following screenshots of the actual application:

1. Main dashboard page.
2. Crop recommendation output.
3. Crop-yield prediction output.
4. Market-price prediction output.
5. Weather-risk analysis output.
6. Agricultural data visualization.
7. Model evaluation results.

7.5 User Manual

1. Open the AgriIntelligence application.
2. Navigate to the required agricultural module.
3. Provide the required input parameters or select the available dataset.
4. Submit the inputs for analysis.
5. View the generated predictions and visualizations.
6. Interpret the results to understand agricultural trends and potential risks.